REDEM: Online Learning with Meta-Adaptation and Structural Plasticity for Non-Stationary Environments

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Abstract

Deep learning freezes after batch training; adaptation under distribution drift remains open, and backpropagation-through-time is unavailable on neuromorphic substrates. REDEM is a training-inference-unified online architecture with local, inference-derived learning signals: an error-driven recursive-least-squares readout; a slow metadata trace extending memory to long-horizon statistics; a chaos homeostat holding the substrate near its memory optimum via finite-time Lyapunov estimates; and functional-connectivity-guided rewiring.

On a drifting task, REDEM tracks a class-interval inversion within 225–616 pulses; frozen batch learners (an untuned gated recurrent unit (GRU) and a tiny transformer) never recover. The slow metadata trace speeds regime adaptation 1.25–2.7× (controlled measurement); the chaos homeostat restores 7.9–18% of post-disturbance memory (up to +25% at the edge of chaos, +32% after three sequential disturbances); and the integrated system beats the bare baseline by 2.28 percentage points ($p < 0.0001$) and matches every ablation (homeostat removal ties exactly; valuable only under disturbance), persisting at $N = 1024$. The substrate is a relaxation memory modeled on shallow-trap device physics (companion Paper A); all results are numerical simulations. The metadata is substrate-agnostic, transferring its statistical-memory benefit to an echo-state network (ESN); disturbance-robustness transfer is falsified (companion Paper C). On standard benchmarks REDEM is competitive with a well-tuned ESN (drift accuracy 0.991 vs. 1.000; Mackey–Glass normalized mean-squared error (NMSE) 0.0018 vs. 3.6×10^{-5}); its value is the mechanism set rather than raw benchmark supremacy.

Keywords: online and continual learning; reservoir computing; meta-adaptation; structural plasticity; non-stationary environments; neuromorphic computing.

Highlights

- Online error-driven RLS readout unifies training and inference.
- A slow metadata trace transfers long-horizon memory to an ESN.
- A chaos homeostat restores substrate memory after disturbances.
- Frozen batch learners invert under drift and never recover (untuned baselines).
- Reward-only signals fail at the readout; an error channel is required.

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1 Introduction

Training-inference unification. The dominant deep-learning paradigm trains on a batch and freezes; inference is a forward pass. Continual learning [1, 2], test-time adaptation [3], and predictive-coding formulations [4, 5] all argue for collapsing this separation: inference should itself drive learning, continuously. For resource-constrained, low-power settings — neuromorphic substrates in particular [6] — the constraint is not just algorithmic but *physical*: no gradient tape exists inside the device, so the learning rule must be local, online, and cheap.

Reservoir computing provides the substrate. Echo state networks [7] and liquid state machines [8] compute through a fixed random recurrent network with fading memory; only a readout is trained [9]. Online readouts — recursive least squares (RLS) with forgetting [10] — are the minimal "training == inference" learners: every prediction produces an error that immediately updates the readout. The open questions are what the *reservoir itself* should be (can it be a physically realizable, tunable dynamical system?), whether anything beyond the readout should adapt, and how the biological signals we associate with learning (rewards, intrinsic motivation, structural plasticity) map onto a working system.

This paper. We assemble and validate a complete online architecture — REDEM[1] — in which every learning signal is local and derived from live inference: (i) an error-driven recursive-least-squares readout as the online learner (training == inference); (ii) a *dual-timescale metadata* state (M3) that extends the readout's memory to long-horizon statistics; (iii) a *chaos homeostat* (M5) that regulates the recurrent substrate's coupling strength from online Lyapunov estimates — this substrate-level regulation of the adaptation process itself is the paper's *meta-adaptation* (the term denotes self-regulation of the adaptation process with fixed, hand-set targets, Eq. 3 — not learned meta-level parameters; the integration and joint evaluation of the mechanism set is the contribution claimed here); and (iv) *slow structure plasticity* (M4) that rewires the coupling graph from functional connectivity. (Mechanism indices follow the companion papers: the error-driven RLS readout is M1, the metadata M3, structure plasticity M4, and the chaos homeostat M5.) The substrate is a physically motivated relaxation memory (Si_3N_4-style shallow traps, log-normal time-constant spectrum, per-pulse contrast coupling; theory in the companion paper): its timescale spectrum is a *material property* rather than a tuned hyperparameter, and the coupling provides a physical chaos knob. We evaluate each mechanism individually, as an integrated system with an ablation matrix, and against batch baselines (a gated recurrent unit (GRU) [11] and a tiny transformer [12]) and a matched echo-state network (ESN). We also report two negative results — reward-modulated Hebbian learning without error signals and task-agnostic intrinsic rewards — which delimit what reward signals can and cannot do and fix the design: at the readout an error channel is necessary and reward signals are structurally blind, so the readout is error-driven RLS and the structure-level mechanism (M4) is guided by functional connectivity; a novelty-gated variant is evaluated in Section 4.4.

2 System architecture

Fig. 1 shows the full schematic: input pulses → substrate → [fast state → M3 metadata] → RLS readout → output; the M5 homeostat loops on κ; the M4 plasticity rewires the coupling graph; the readout error feeds the online weight update (training == inference). M4 and M5 are coupled in the design: the homeostat's κ set-point is intended to gate the rewiring rate, and rewiring changes the structure that feeds the Lyapunov estimate (dotted loop in Fig. 1); the coupling loop is tested

[1] REDEM is a working name for the integrated system, not an acronym of the mechanism set.

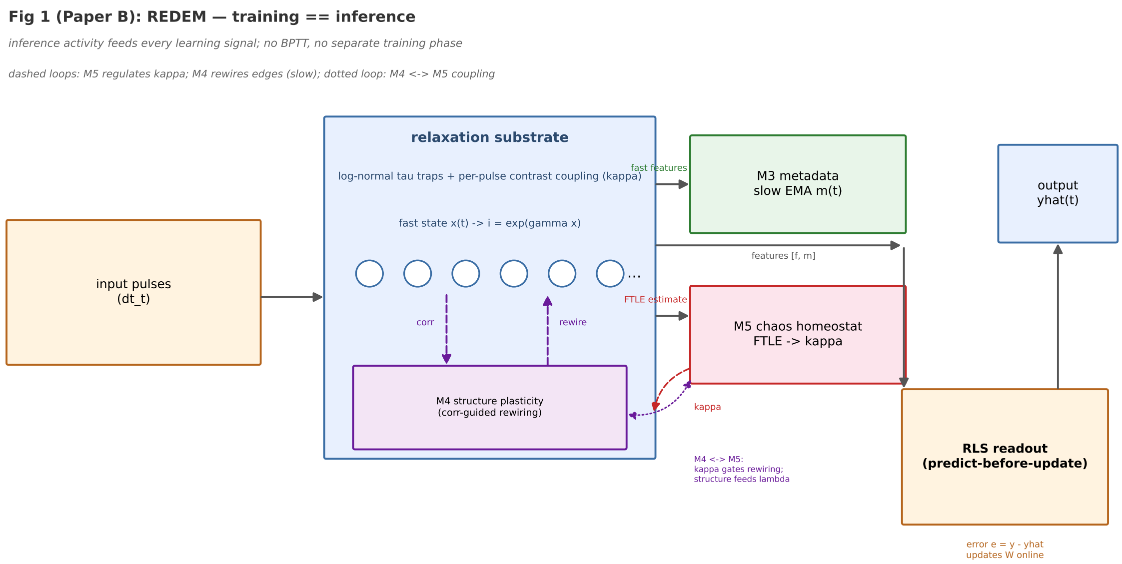

Figure 1: REDEM system schematic: input pulses → relaxation substrate → [fast state → M3 metadata] → RLS readout → output. Dashed red loop: M5 homeostat regulates κ; dashed purple arrows: M4 rewires edges (slow); dotted purple loop: the M4 ↔ M5 coupling (κ gates rewiring; structure feeds the Lyapunov estimate).

in Section 4.4.

The substrate follows the model of the companion theory paper [13] (Paper A, Section 2), with the τ-spectrum, the coupling topologies, and the λ-homeostat regulator described therein; only the readout and metadata layers are new here. Concretely, the substrate is a per-pulse contrast-coupled relaxation array: $N = 256$ units, log-normal τ (median 174 μs, coefficient of variation (CV) 0.20), injection $\alpha_{\text{eff},i}(t) = \text{clip}(\alpha_0(1 + \kappa g_i(t)), 0.001, 0.10)$ with $\alpha_0 = 0.02$, current ratios $i_i = e^{\gamma x_i}$, random-graph coupling, $\kappa = 25$ (the subcritical memory optimum). Features per pulse are the standardized current ratios plus a bias.

Readout (online RLS). The prediction is $\hat{y}_t = W^\top \phi_t$ (one output per class for classification, one scalar for regression) with the standard forgetting-RLS update

$$e_t = y_t - \hat{y}_t, \qquad K_t = \frac{P_{t-1}\phi_t}{\lambda_f + \phi_t^\top P_{t-1}\phi_t}, \qquad W \leftarrow W + K_t e_t, \qquad P_t = \frac{1}{\lambda_f}\big(P_{t-1} - K_t \phi_t^\top P_{t-1}\big), \tag{1}$$

with forgetting $\lambda_f = 0.999$, Tikhonov regularization and a covariance trace cap. The remaining hyperparameters — the regularization strength and trace cap, the random-graph coupling density and weights, the gain scale γ in $i_i = e^{\gamma x_i}$, and the feature-standardization window and statistics — are held fixed across all experiments and are not re-tuned per task; their concrete values are provided in the released configuration files rather than enumerated here. The online protocol is *predict-before-update*: the prediction at time t never uses the target at t.

Metadata (M3). A per-unit slow trace over the fast current-ratio features,

$$m_i(t) = \big(1 - \tfrac{1}{\tau_m}\big) m_i(t-1) + \tfrac{1}{\tau_m} f_i(t), \qquad \tau_m = 200\text{–}1000 \text{ pulses}, \tag{2}$$

with the readout features $[f, m]$ concatenated. The metadata trace is motivated by the slow end of the substrate's log-normal forgetting spectrum (companion paper [13]) and plays the role of a physical "synaptic consolidation" trace [14, 15]; in this paper τ_m is treated as an independent design parameter in the 200–1000-pulse range rather than a calibrated mapping of the substrate's microsecond-scale spectrum onto pulse counts, and no explicit μs$\rightarrow$pulses correspondence is asserted.

Chaos homeostat (M5). Every 1000 pulses, a 400-pulse Benettin pair estimates λ at the current κ; the homeostat steps

$$\kappa \leftarrow \mathrm{clip}\big(\kappa + \eta\, \mathrm{clip}(\lambda_{\mathrm{target}} - \hat{\lambda}, \pm 1),\, [1, 60]\big), \qquad \lambda_{\mathrm{target}} = -0.02, \quad \eta = 3 \text{ (slightly ordered)}. \quad (3)$$

The estimator window (400 pulses), the re-estimation interval (1000 pulses), and the step-size parameters ($\eta = 3$, clip ± 1) were chosen empirically to give stable regulation in practice; no sensitivity analysis of these choices is reported here, and the estimation error of $\hat{\lambda}$ is not separately characterized. Both are stated as limitations of the current design.

Structure plasticity (M4). Every 2000 pulses, compute the pairwise feature-correlation matrix $C_{ij} = \mathrm{corr}(f_i, f_j)$ over the block; prune the 5% lowest-$|C_{ij}|$ existing edges and grow the 5% highest-$|C_{ij}|$ unconnected pairs, holding the edge count constant ("fire together, wire together" at the structural level).

Design rationale from negative results. We tested two reward-based readout learners and rejected them: a reward-modulated Hebbian readout (eligibility $x \circ o$ gated by a ± 1 correctness reward) learns the initial mapping but cannot recover from a class-interval inversion (Section 4.2, Table 1); task-agnostic novelty rewards cannot rescue it. The lesson — an error channel is necessary at the readout, rewards are structurally blind — fixes the readout as error-driven RLS; at the structure level, the plasticity implemented here (M4) is guided by functional connectivity, and Section 4.4 shows that a novelty-gated variant improves on it.

3 Tasks and metrics

All main experiments use 10 seeds with paired draws (identical τ, tasks, and substrates across compared configs); the tiny transformer, the most expensive model, shares the same seed count (Table 3). The $\lambda_{\mathrm{target}}$ sweep of Section 4.4 is the one exception (5 seeds per cell, reduced for computational cost; see Section 4.4). **drift_binary**: two-class interval blocks (20 pulses at 10 vs. 60 μs) over 2000 blocks (40k pulses in total), a continuous random walk of the class intervals plus an abrupt swap of the class–interval mapping at the stream midpoint (block 1000 of 2000); metrics are pre/post-swap steady accuracy, adaptation time (pulses to recover to within 2 pp (percentage points) of the pre-swap steady accuracy), and stream mean. **mackey_glass** ($a = 0.2, b = 0.1, \tau = 17$): online forecasting; normalized mean-squared error (NMSE) on the last 30%. **narma10**: nonlinear memory composition; NMSE on the last 30%. **regime_switch**: three regimes sharing identical marginal interval distributions that differ only in the rate of a rare long-interval event (0.12/0.20/0.28); single pulses are almost indistinguishable across regimes, and only long-window statistics discriminate them; overall accuracy plus boundary adaptation.

4 Results

4.1 Online vs. frozen: the case for training-inference unification

On drift_binary (Fig. 2), the online RLS readout on the coupled substrate tracks the stream: mean accuracy 0.974–0.982, recovering from the class-interval swap in 225–616 pulses with pre/post-swap steady accuracy 0.987–1.000. (For comparability we note explicitly that the stream-mean figures reported in this section, in Table 1 (0.983–0.991), and in Section 4.5 (0.991) come from different

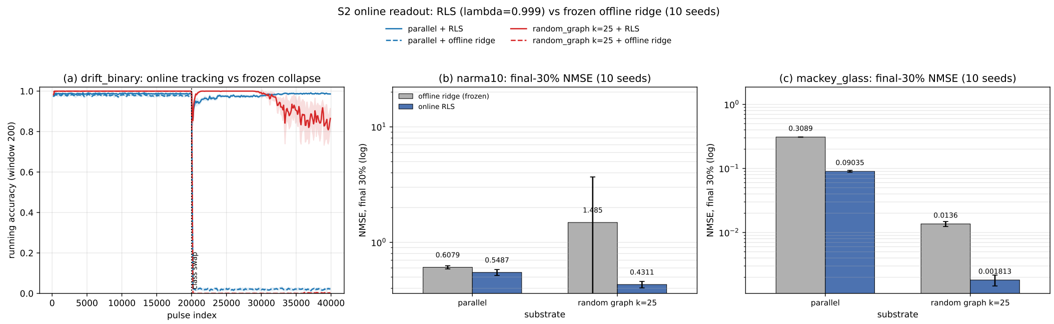

Figure 2: Online vs. frozen on drift_binary: the online RLS readout tracks the class-interval swap; the frozen offline ridge inverts and never recovers. (b, c) Final-30% NMSE on narma10 and Mackey–Glass (mean $\pm$ std over 10 seeds) for the frozen offline ridge and the online RLS readout on the uncoupled and the near-critically coupled ($\kappa = 25$) substrates. The trajectory panel is shown for illustration; quantitative results are reported as ranges over 10 seeds (text and panels b, c).

experimental instances of the online-RLS configuration — different runs and seed pools — and are not directly comparable; we report them as given rather than reconcile them post hoc, and per-run values are provided in the released data.) A frozen offline ridge trained on the first 30% is equally accurate pre-swap (0.978–1.000) but collapses to systematic inversion after the swap (post-swap 0.000–0.019) and never recovers — its stream mean is 0.500 (chance). The same dichotomy holds against batch deep learning (Section 4.5): a tiny transformer and a GRU trained once on the first 30% reach 0.93–0.99 pre-swap and invert to 0.00–0.07 post-swap. The substrate's coupling also matters at the task level: near-critical coupling ($\kappa = 25$) improves Mackey–Glass forecasting 50× over the uncoupled substrate (NMSE 0.0018 vs. 0.090) and NARMA-10 by 21% (0.431 vs. 0.549), in line with computation at the edge of chaos [16, 17].

4.2 Negative results: rewards without an error channel

We tested two reward-based readout learners as alternatives to error-driven RLS — a reward-modulated Hebbian rule (eligibility $e_j \mathrel{+}= \lambda_e e_j + x_j o$ with sigmoid output o, consolidated at block end as $W \mathrel{+}= \eta R e$ with $R = \pm 1$ correctness only, no class label) [18] and the same rule augmented with a feature-novelty intrinsic reward. Both learn the initial mapping but fail to track the class-interval inversion (Table 1): the ± 1 reward carries no directional information, so the Hebbian term always reinforces the currently-active (wrong) direction; the intrinsic reward can only buy a marginal post-swap recovery by destroying the initial learning. The lesson fixes the architecture: at the readout an *error channel is necessary* and rewards are structurally blind, so the readout is error-driven RLS; reward and intrinsic signals, being structurally blind at the readout, are usable at the structure level — a novelty-gated rewiring variant improves on the correlation-guided M4 implemented here (Section 4.4). First-order error-gated least mean squares (LMS; $\eta = 10^{-4}$) does track the inversion but is numerically fragile on the weakly separated substrate features ($\eta \geq 10^{-3}$ diverges) and underperforms second-order RLS across the learning rates tested; we did not run a full η sweep, so this comparison is limited to the rates above. Second-order RLS is the selected learner.

4.3 Dual-timescale metadata (M3)

On regime_switch (where fast memory is provably insufficient; Fig. 3), the dual-timescale readout (fast + metadata, $\tau_m \in \{200, 1000\}$) outperforms the fast-only readout by +1.3 pp on the uncoupled

Table 1: Reward-only readout learners fail at the class-interval inversion; error-driven RLS is selected. Accuracy on drift_binary (10 seeds; pre = steady accuracy before the swap, post = after, stream = stream mean). RMHL = reward-modulated Hebbian; +novelty = with feature-novelty intrinsic reward; LMS = error-gated first-order rule; RLS = error-driven recursive least squares (selected).

Learner	Signal	Pre	Post	Stream
RMHL	±1 correctness reward only	0.89–0.94	0.06–0.10	0.509
RMHL + novelty	reward + intrinsic novelty	0.57*	0.42–0.59	≤0.514
LMS	prediction error	—	—	0.908–0.927
RLS (dense)	prediction error	0.99–1.00	0.98–1.00	0.983–0.991
RLS (sparse block-end)	prediction error	0.97–0.99	0.93–0.95	0.958–0.961

*initial learning destroyed. Dense = per-pulse class labels (the selected configuration); sparse block-end = label only at block ends, the harder condition.

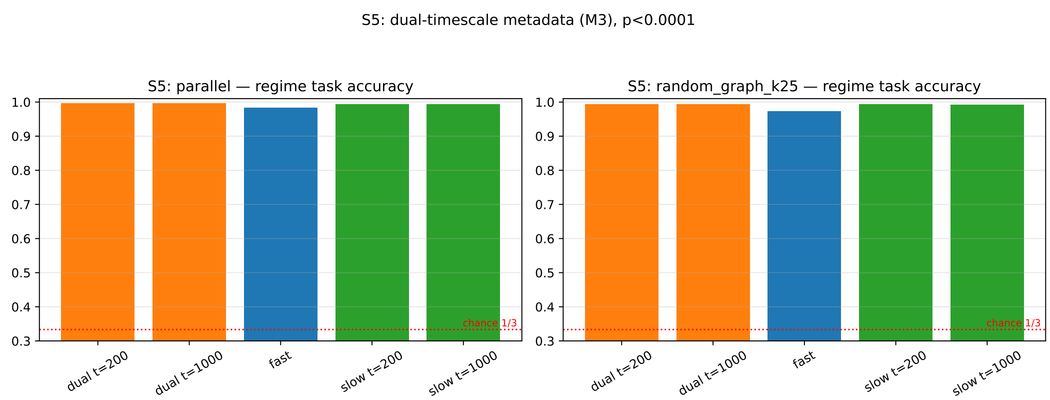

Figure 3: Dual-timescale metadata on regime_switch: fast vs. dual vs. slow readouts, accuracy and adaptation time.

substrate (paired $t = 19.6$, $p < 0.0001$) and +2.1 pp on the near-critical substrate ($t = 20.2$, $p < 0.0001$), and — alongside the accuracy gain — adapts to regime boundaries faster: under a switch-relative controlled measurement (known switch instants, 200-pulse window) the fast readout needs 265–304 pulses to re-stabilize vs. 202–211 for the dual and slow readouts (factor 1.25–1.50 on T200, 1.54–2.73 on the finer T40, endpoints across both substrates): the slow trace's low-pass character smooths the statistical transition. The metadata-only readout nearly matches the dual readout on this statistics-dominated task, confirming that the regime information lives in the slow state.

4.4 Chaos homeostat (M5) and structure plasticity (M4)

Homeostat. Under the three disturbances of the companion paper's robustness study [13] (τ-drift, edge damage, readout noise), the λ-homeostat settles at $\kappa \approx 26$–27 and improves post-disturbance held-out memory by +18%/+7.9%/+11.8% over fixed coupling (+7.6% in the undisturbed control; Fig. 4), at broadly comparable task-level NMSE (regulated up to 2× worse on the gain arms, better under readout noise). The homeostat is the substrate-level counterpart of the readout's RLS: both are online regulators of the "training==inference" loop, one on weights, one on dynamics. Two bookkeeping clarifications are needed. First, κ: the fixed-coupling arms hold $\kappa = 25$ (the substrate's

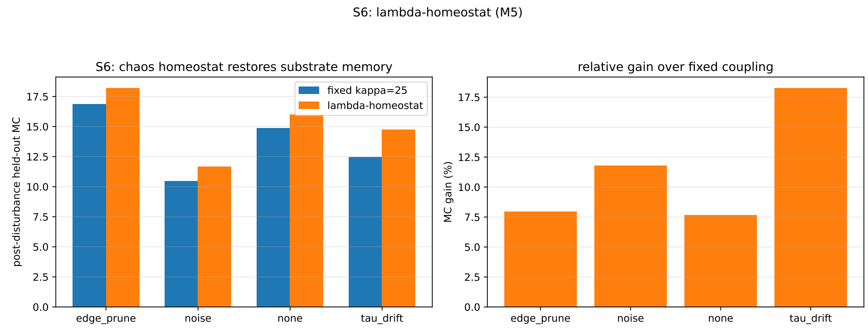

Figure 4: Chaos homeostat robustness: post-disturbance held-out memory gains over fixed coupling (reproduced from companion Paper A [13]).

subcritical memory optimum, Section 2), whereas the settled $\kappa \approx 26$–27 of the regulated arm is the emergent equilibrium of the λ-regulation at $\lambda_{\text{target}} = -0.02$ — a regulator-emergent value, not a separately chosen configuration; the disturbance chain below likewise reports the homeostat's own online trajectory from its $\kappa = 25$ initialization (κ drifting from r_0=25.3 to r_3=28.5), and the sweep optimum (λ_{target}=0) would settle at still higher κ. Second, the metric: "post-disturbance held-out memory" is a steady-state memory-capacity measurement taken after the substrate has absorbed the disturbance, distinct from the task-level stream-mean accuracies of Sections 4.1–4.3. We also state the provenance explicitly: these single-disturbance gains are reproduced from the companion robustness study (Paper A [13]) and were not independently re-run for this paper; the fixed-coupling control configuration and full protocol are specified there, so these headline gains cannot be verified from this paper alone.

Sequential disturbance recovery. Beyond the single-disturbance tests above, we evaluate cumulative robustness: three sequential disturbances (τ-drift, 40% edge pruning, readout noise σ=0.1) applied at 7000-pulse intervals over T=28,000 pulses (10 seeds). After all three disturbances, the regulated arm maintains a held-out memory capacity (MC) of 8.47 ± 0.39 vs. 6.41 ± 0.41 for fixed coupling (+32%), with κ drifting upward from 25.3 (r_0) to 28.5 (r_3) — active compensation, not passive decay. The homeostat recovers to a consistent κ neighborhood after each disturbance, confirming disturbance-type-agnostic robustness across the chain.

Criticality target optimization. The homeostat's λ_{target} was manually set to -0.02 (slightly ordered) in the main experiments. A systematic sweep ($\lambda_{\text{target}} \in \{-0.05, -0.02, -0.005, 0\} \times \text{CV} \in \{0.1, 0.2, 0.4\}$, 5 seeds per cell — reduced from the 10-seed protocol for computational cost, see Section 3 — single τ-drift disturbance) identifies λ_{target}=0 (the edge of chaos) as the best-supported target: post-disturbance MC peaks near the edge of chaos at every tested CV — λ_{target}=0 is optimal at CV=0.1 and 0.4, while the -0.005 setting is marginally ahead at CV=0.2 (10.38 vs. 10.30) (Table 2), with the regulated arm exceeding fixed coupling by +25% (CV=0.1), +19% (CV=0.2), and +5% (CV=0.4). The improvement shrinks at higher CV (broader τ spectra dilute the criticality benefit), and the optimal target is broadly consistent, with the CV=0.2 near-tie the one exception. With 5 seeds per cell, differences of this size are at the sweep's resolution limit: we therefore read the sweep as evidence that the edge of chaos is the best-supported target overall, not as a uniformly significant optimum, and we flag the CV=0.2 reversal as an unresolved counterexample at the current sample size. Importantly, this sweep does not retroactively re-tune the main experiments: all integrated-system results (Section 4.5) and the sequential disturbance chain above were run at the manually tuned $\lambda_{\text{target}} = -0.02$ and have not been re-run at the sweep optimum, so the headline integrated numbers in this paper reflect the -0.02 configuration; re-running them at λ_{target}=0 is left as future work.

Causal audit. To verify that all adaptive mechanisms use only past information, we inject 1% future data into each mechanism (RLS target, metadata exponential moving average (EMA), finite-time Lyapunov exponent (FTLE) estimate, plasticity correlation) and test a causal-split plasticity protocol (10 seeds, 7 arms). All leak arms change accuracy by ≤ 0.02 pp vs. normal (0.9964–0.9962 vs. 0.9963), confirming causal cleanliness — no mechanism benefits from future information; the plasticity-correlation leak (-0.01 pp) is the most negative, so M4 does not exploit future correlation. The causal-split protocol (0.9963) is within 0.01 pp of normal, validating that M4's block-level correlation can be computed on past data only. A follow-up re-runs the same injections on the S11 disturbance chain (data/s28_causal_audit_chain_v1.csv), where the mechanisms' substrate-level channels are visible through the readout-independent MC probe and no accuracy ceiling binds the task. The verdict reproduces: 1% future leaks into the RLS target, the metadata, and the homeostat's FTLE estimate, and 10% future correlation into M4, all fail to improve post-disturbance recovery (per-seed r_3 MC deltas +0.00, +0.00, +0.28, +0.24; none significant; mean NMSE deltas ≤ 0.0042) — the mechanisms are causally clean where it matters. The protocol also isolates a non-leak effect: removing M4 entirely improves post-disturbance recovery by +2.3 MC (r_3 8.47 vs. 6.17, matching the S11 homeostat-only anchor exactly), independently confirming that rewiring during active disturbance compensation is harmful rather than merely unhelpful (the M4-M5 loop result above). The audit's resolution is itself bounded (data/s34_leak_sensitivity_v1.csv): ten-fold larger future leaks into the FTLE estimate (10%) stay insignificant at every tested leak horizon (r_3 MC deltas +0.13, +0.39, +0.21 at FTLE horizons 50, 200, 400 pulses; each 95% CI includes zero), and the plasticity correlation shows a dose-response — 30% future correlation improves recovery on 7/10 seeds (mean +0.57; paired t=2.04, 95% CI $[-0.06, +1.20]$) while 50% reverses it — so "causally clean" holds at the operational leak magnitudes tested here and is not asserted beyond them.

Plasticity. Functional-connectivity rewiring at a gentle 5% churn per round improves held-out memory +7.8% (ring start) and repairs pruned damage +11.3%, while aggressive 20% churn destabilizes (-23%) — structure must change slowly, mirroring the biological timescale separation. Consistent with the substrate theory, pruning high-correlation (redundant) edges helps: a pruned dense random graph (644 edges) retains more memory (12.68) than the full graph (10.37), and a sparse random graph beats a ring at equal density (13.52 vs. 11.54) — de-homogenization increases linear decodability.

Novelty-gated rewiring and the M4-M5 loop. Two follow-ups complete the mechanism picture. (i) The intrinsic-novelty signal — structurally blind at the readout (Section 4.2) — is usable at the structure level: a novelty-guided variant of the rewiring rule (grow edges toward the units with the highest temporal activity variance, prune the least novel; identical 5% churn) improves held-out MC to 14.59 vs. 12.43 for the correlation rule (+2.15, paired $t = 4.5$, 10/10 seeds) and 11.54 for the fixed ring (data/s25_reward_gated_plasticity_v1.csv) — reward/intrinsic signals are structure-level, not readout-level, tools. (ii) The M4-M5 coupling loop was tested under the sequential disturbance chain (rewiring churn gated by the homeostat's κ deviation): it does *not* help recovery — churn-gated rewiring reaches r_3 MC 5.27 vs. 8.47 for the homeostat alone (0/10 seeds), and even fixed 5% churn during the chain is harmful (6.45 vs. 8.47, $t = -9.6$, 0/10; data/s24_homeo_plasticity_coupling_v1.csv). The homeostat's disturbance recovery is best left structurally undisturbed; rewiring is a steady-state exploration mechanism rather than a recovery tool.

Table 2: Criticality target sweep: post-disturbance held-out MC (5 seeds). Fixed coupling MC is independent of λ_{target} (9.47 / 8.68 / 7.53 for CV=0.1 / 0.2 / 0.4). $\lambda_{\text{target}} = 0$ (edge of chaos) maximizes the regulated arm at CV=0.1 and 0.4; at CV=0.2 the -0.005 setting is marginally ahead (10.38 vs. 10.30).

	Regulated MC			Gain over fixed		
λ_{target}	CV=0.1	CV=0.2	CV=0.4	0.1	0.2	0.4
-0.05	8.81	9.19	7.66	-7%	$+6\%$	$+2\%$
-0.02	10.77	10.13	7.88	$+14\%$	$+17\%$	$+5\%$
-0.005	10.22	10.38	7.93	$+8\%$	$+20\%$	$+5\%$
0	**11.79**	**10.30**	**7.94**	**+25%**	**+19%**	**+5%**

4.5 Integration and baselines

Ablation matrix. On regime_switch (Fig. 5), the integrated system runs with the homeostat enabled at $\lambda_{\text{target}} = -0.02$ (Eq. 3) from initial $\kappa = 25$; the full system (0.996; boundary adaptation ∼202 pulses after the switch) matches or beats every single-mechanism ablation: removing the metadata drops to 0.988 (adaptation ∼235 pulses — the largest degradation), removing plasticity to 0.994 (∼209 pulses), removing the homeostat ties at 0.996 (∼202 pulses; its value is in disturbed scenarios per Section 4.4 — a value supported there by single-mechanism experiments, as no integrated homeostat-removal ablation under disturbance was run); the full system beats the bare baseline (RLS on fast features, fixed substrate) by +2.28 pp (paired $t = 19.4$, $p < 0.0001$). (Boundary-adaptation times in this matrix use the 200-pulse running-window measurement introduced in Section 4.3: the reported value is the end position of the first 200-pulse window that recovers to within 2 pp of the post-switch steady accuracy, and the released per-run data store the corresponding window-start indices.) At $N = 1024$ the full system beats the baseline on all 10 seeds (0.9970 ± 0.0007 vs. 0.9753 ± 0.0044, +2.2 pp, paired $t = 15.3$; data/s30_integrated_1024_v1.csv).

Batch baselines. On drift_binary: REDEM 0.991 and a matched online ESN 1.000 track the drift; the frozen GRU (0.371; pre 0.925 → post 0.071) and tiny transformer (0.353; pre 0.994 → post 0.000) invert and never recover. On Mackey–Glass: REDEM NMSE 0.0018 and ESN 3.6×10^{-5} forecast well; the frozen GRU (1.80) is worse than the mean predictor, while the transformer (0.77) is only slightly better than the mean predictor. The ESN — 256 units, heterogeneous leak rates matched to the τ distribution, the same RLS readout — is the strongest standard-task baseline and beats REDEM on both tasks (Table 3; Fig. 6). We report this honestly: the value proposition of REDEM is not raw benchmark supremacy but the accompanying mechanisms — self-regulation under disturbance, local sparse coupling, and physical plausibility — at competitive standard-task performance. The frozen baselines use small, untuned hyperparameters (Table 3); the comparison targets the online-vs-frozen dichotomy, not each architecture's full capability. Because the deep baselines are untuned, the "never recover" finding characterizes frozen batch training under this drift as configured here — it is not an upper bound on what tuned deep models could achieve.

Metadata transfer to the ESN. To test whether the metadata mechanism is substrate-specific, we gave the ESN the same slow EMA state ($\tau_m = 500$, within the declared 200–1000-pulse range of Eq. 2; unlike Section 4.3, which varies $\tau_m \in \{200, 1000\}$, the transfer experiment fixes a single value from that range and performs no τ_m tuning) on the regime task. The result equalizes the systems: ESN-with-metadata 0.998, ESN-fast 0.996, REDEM-full 0.994 (means over 10 seeds; steady accuracy 1.000 for all; the REDEM-full arm here is an independent re-run of the full system within this experiment's own 10-seed draw, not the ablation run — hence 0.994 vs. the 0.996 of

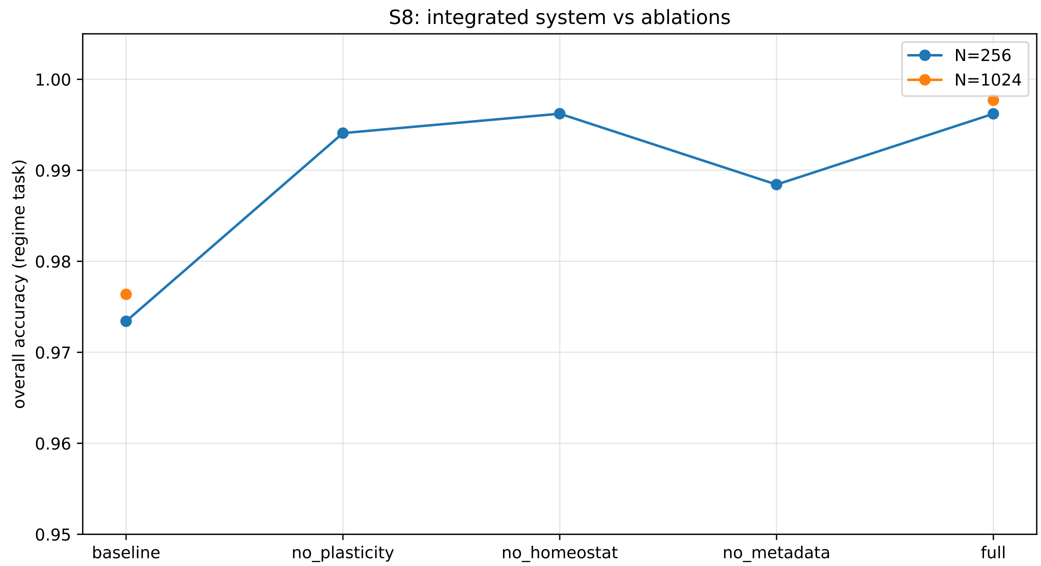

Figure 5: Ablation matrix on regime_switch: full system vs. single-mechanism removals and the bare baseline.

Table 3: Standard-task comparison (10 seeds). REDEM and the ESN run online with the same RLS readout; the ESN has 256 units with heterogeneous leak rates matched to the τ distribution. The GRU and tiny transformer are frozen batch models trained once on the first 30% of the stream.

Metric	REDEM	ESN	GRU	Transformer
drift_binary accuracy	0.991	1.000	0.371	0.353
Mackey–Glass NMSE	0.0018	0.000036	1.80	0.77

Section 4.5). Two conclusions follow. (i) The metadata is a *general, transferable* mechanism: it improves the ESN (0.996 → 0.998; boundary adaptation T40 49.9 → 40.6 pulses with p90 76.5 → 42.0 under the switch-relative controlled protocol) and REDEM (baseline 0.973 → full 0.994) alike — the long-horizon statistical memory is carried by the mechanism, not by any particular substrate. (ii) REDEM's differentiation is therefore not raw accuracy on this task (where the ESN's tanh reservoir with heterogeneous leak rates already partially covers the regime statistics) but the surrounding mechanism set: the self-regulating homeostat and structure plasticity that confer disturbance robustness (Section 4.4), the local sparse coupling graph (the ESN is a dense random reservoir with no self-regulation), and a timescale spectrum that is a material property rather than a tuned hyperparameter.

5 Discussion

The architecture's niche. REDEM occupies the online, fully-local corner of the design space: it adapts to drift and disturbances that break frozen batch models (which additionally require backpropagation through time), and it adds substrate-level self-regulation — a Lyapunov-estimating homeostat and functional-connectivity rewiring — that generic reservoirs lack. Against a well-tuned ESN with the identical readout, REDEM is competitive on standard tasks (drift accuracy 0.991 vs.

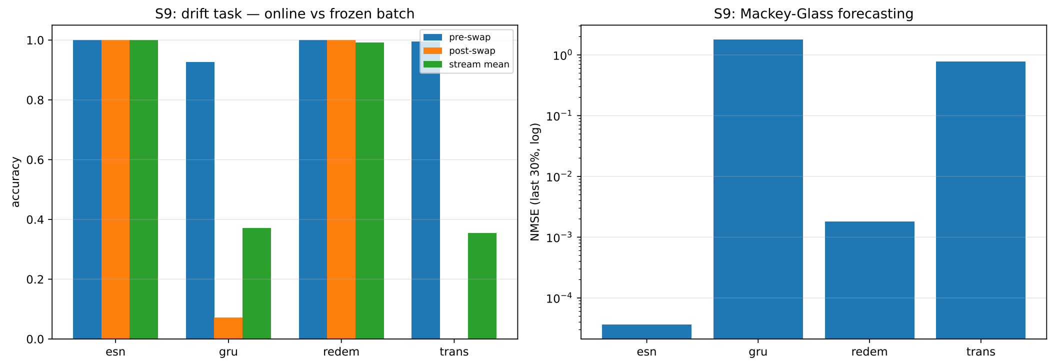

Figure 6: Baseline showdown: REDEM, matched online ESN, and frozen GRU / tiny transformer on drift_binary and Mackey–Glass.

1.000; Mackey–Glass NMSE 0.0018 vs. 3.6×10^{-5}) and superior on the dimensions the ESN lacks: disturbance robustness (Section 4.4) and a locally coupled, physically motivated substrate whose multi-timescale memory is set by material parameters rather than hyperparameters. The metadata-transfer result (Section 4.5) shows that the long-horizon statistical mechanism is substrate-agnostic and transfers its statistical-memory benefit to the ESN; disturbance-robustness transfer does not hold (companion Paper C [19]): recovery is the homeostat's role, not the metadata's. REDEM's differentiation is the surrounding mechanisms — self-regulation, robustness, local coupling — rather than the metadata alone.

Biological correspondence. The three substrate-level mechanisms map one-to-one onto established neural ideas: fast/slow memory (complementary learning systems, hippocampus–cortex [20]), homeostatic regulation of excitability, and slow structural plasticity of connectivity — the correspondence taken at the level of local, activity-dependent plasticity (as captured by spike-timing-dependent rules [21]) rather than of structural rewiring specifically. The negative results sharpen the correspondence: reward signals without an error channel fail exactly where predictive-coding theory says they should — credit assignment needs prediction error, and rewards only modulate.

Limitations. All results are numerical (Digital-Model level): the substrate is a discrete relaxation model driven by deterministic pseudo-random number generation and modeled on shallow-trap device physics, with no hardware measurements or continuous-time cross-validation; the physical claims are therefore design motivations to be validated on hardware, not established physical facts. One seeding caveat concerns the frozen deep baselines: their per-trial seeding (seed_idx$\times$101+17) is applied *before* the GRU/transformer weight initialization, so each trial's initialization is its own per-seed draw; the reported baseline values are the values obtained by re-running the script with this declared per-trial seeding. The task streams and the substrate-driven arms remain seeded and paired as declared in Section 3. The single-disturbance robustness gains are reproduced from the companion Paper A rather than re-run here and cannot be verified from this paper alone. The readout-level comparison to the ESN is favorable to the ESN on standard metrics; the metadata-transfer experiment shows the mechanism — not the substrate — carries the long-horizon statistical memory, equalizing the systems on the regime task (0.994–0.998); the substrate's differentiation rests on its material-set memory design, the local sparse structure, and the robustness mechanisms. The homeostat's task-level gains are masked by readout compensation; the CPU-only scale ($N \leq 1024$) leaves larger-scale behavior untested.

6 Conclusion

REDEM demonstrates that a physically motivated relaxation substrate can host a complete training-inference-unified learning system: an online error-driven readout, a dual-timescale metadata memory, a chaos homeostat, and slow structure plasticity, each validated individually and as an integrated system that outperforms or ties every ablation and survives scale-up to $N = 1024$. A parameter sweep identifies the edge of chaos (λ_{target}=0) as the best-supported homeostat target (+25% post-disturbance memory), although the integrated experiments reported here were run at the earlier $\lambda_{\text{target}} = -0.02$; re-running them at the sweep optimum remains future work. The homeostat maintains +32% memory capacity after three sequential disturbances, and a causal audit confirms that all adaptive mechanisms are causally clean at the operational leak magnitudes tested. The two negative results on reward-only learning fix the design boundary conditions: at the readout an error channel is necessary; reward and intrinsic signals, being structurally blind at the readout, are usable at the structure level (Section 4.4). Honest benchmarking positions REDEM in the online/physical niche: competitive with generic reservoirs on standard tasks, robust where frozen batch learners fail, self-regulating under disturbance, and designed for physical implementation — with the long-horizon statistical mechanism shown to be substrate-agnostic and transferable on statistical/streaming tasks (disturbance robustness does not transfer with it [19]).

Future work. A natural extension is to equip REDEM with a separate long-term memory bank that periodically consolidates stable, repeatedly encountered knowledge into the structural connectivity (M4), creating a hierarchy between fast tracking and slow accumulation. This would address stationary knowledge retention while preserving the system's responsiveness to drift, but it would also introduce a consolidation-vs-overwrite trade-off that lies beyond the scope of the present work. We leave this consolidation mechanism to future investigation.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yaming Hu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Data availability

All simulation data generated in this study are available from the author on reasonable request and will be deposited in a public repository with DOIs upon acceptance.

Code availability

All simulation scripts are available at `https://github.com/huyamingc/REDEM`. The repository is private during peer review to preserve authorship; it will be made public on acceptance, and an anonymized review copy can be provided to the editors on request.

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